

HOTEL MANAGEMENT SYSTEM

SYSTEM PROPOSAL

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ABSTRACT

The Hotel Management System is a software solution designed to make work easy and automate the daily operations of a hotel. This system aims to improve and make easy activities such as room booking, guest check-in and check-out, billing, staff management, and record-keeping. Traditional methods of hotel management are usually manual, which are time consuming, easy to get errors, and not able to handle large volumes of data. The proposed system solves these challenges by offering an integrated system that ensures real-time access to guest information, room availability, and financial transactions.

The development of this system involved finding the important requirements of hotel staff, administrators, and customers, and turning them into functional and non-functional specifications. The system is designed with a user-friendly interface, securing data, and scalability to accommodate future growth. By automating routine tasks, the Hotel Management System reduces workload, minimizes human errors, and improves customer service delivery.

This project is important for modern hotel operations, where speed, accuracy, and customer satisfaction are mostly needed. The implementation of such a system not only improves just efficiency but also gives hotels an advantage in the industry.

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DEFINITION OF KEY TERMS

Hotel Management System (HMS): A software application designed to automate and manage hotel operations such as room booking, check-in/check-out, billing, staff coordination, and kitchen orders.

Reservation: The process of booking a room or service in advance, either online or at the hotel's reception.

Billing System: A module in the HMS responsible for generating invoices, managing payments, and maintaining financial records.

Check-in/Check-out: The process of a guest registering upon arrival (check-in) and finalizing their stay and leaving the hotel (check-out).

Guest Profile: A digital record that includes a customer's personal information, preferences, and booking history.

System Security: Measures taken to protect the hotel system from unauthorized access and to ensure data privacy and integrity.

Kitchen Order Tracking: A system feature that monitors and records food orders from guests in real time, ensuring timely delivery and accurate billing.

Real-Time Processing: The immediate capture and display of data or actions, such as live room availability or active food orders.

Cybersecurity: The practice of protecting systems, networks, and programs from digital attacks, ensuring the safety of customer and hotel data.

Scalability: The ability of the HMS to grow or expand its features without affecting performance.

ABBREVIATIONS AND ACRONYMS

Abbreviation Full Term

HMS	Hotel Management System
ICT	Information and Communication Technology
UI	User Interface
SDLC	Software Development Life Cycle
DBMS	Database Management System
PHP	Hypertext Preprocessor
HTML	HyperText Markup Language
CSS	Cascading Style Sheets
SQL	Structured Query Language
API	Application Programming Interface
CRUD	Create, Read, Update, Delete
VSC	Visual Studio Code
XAMPP	Cross-platform Apache, MySQL, PHP, Perl
WAMP	Windows, Apache, MySQL, PHP
UI/UX	User Interface / User Experience

CHAPTER ONE: RESEARCH OVERVIEW

1.1 Statement of Problem

Managing hotel reservations, guest information, and billing manually is time-consuming and easy to get errors. Old methods used by hotels often lead to booking conflicts, financial disagreement, and lack of data security. A Hotel Management System will automate reservations, improve customer service, and manage finances, making hotel operations more efficient and secure. Also the kitchen staff who rely on paper forms and phone calls to capture dining orders and room food orders may have miscommunication, delayed meal serving and difficulty in tracking food service income. Managers lack real-time data regarding room status, order progress and financial performance which may lead to making poor decisions and misunderstanding. Furthermore, the lack of cybersecurity practices put sensitive data such as customer information and payment data at risk. These issues show there is a need for a secure and automated Hotel management system to ensure hotel processes run smoothly and data kept safe and secure.

1.2 Study Justification

Having the challenges identified in manual operation of the hotel, this are the justified aims to design a Hotel Management System that addresses these problems. Making hotel reservations automated will reduce booking errors and conflicts, also putting up a system that tracks kitchen order will reduce delay and prevent miscommunication. The system will also make sure finance got by the hotel are secure and accurate. Using cybersecurity features will protect customer information and the hotel finances. By solving this problems the system will make work more efficient, reduce staff work load, improve customer satisfaction and enable managers to have access to real time data which will help them in decision making and supervising the hotel. The project

is more suitable for making hotels more modern and ensuring better service delivery.

1.3 Research Objectives

The aim of this research is to develop a Hotel Management System that makes the hotel operation to be easy and efficient, improve customer experience, protect sensitive information, and improve on financial services. The system will have automated booking, billing, food order and security features to counter the challenges faced by hotel establishments. The objectives stated below will help achieve this goal.

1.3.1 General Objective

To develop an automated and secure Hotel Management System that resolves inadequate service caused by manual booking process, manual billing, outdated guest services and poor food order services.

1.3.2 Specific Objectives

- Improve booking efficiency by using an automated reservation system that reduces errors.
- Improve customer experience through check-in and check-out processes supported by digital records.
- Improve on financial management system by using a secure billing and invoicing system for better using of hotel revenue.
- Improve data security by using cybersecurity measures to protect guest information and payment details.
- Improve on food order and delivery system that will make sure the customers are satisfied.
- To create a real-time kitchen order system that minimizes delay and miscommunication in food service.

1.4 Scope of the Study

This study focuses on solving challenges faced by medium-sized hotels, mostly those having issues such as booking errors, slow service delivery, poor billing systems, and lack of data security. The research will involve analyzing existing systems and design of an automated system. Key focus will be room booking, kitchen order tracking, secure billing system, and data protection. The system will be tested to check on service speed, accuracy, financial control, and data safety.

1.5 Research Questions

- How can automation improve the hotel booking process?
- What security measures can be used to protect customer data in a hotel management system?
- How can the proposed system improve customer service and operational efficiency?
- What are the financial benefits of using an automated billing system in hotels?

CHAPTER TWO: LITERATURE REVIEW

2.1 Introduction to Literature

The hotel industry has grown significantly over the years, requiring efficient systems to manage operations such as reservations, billing and customer service. A Hotel Management System (HMS) is a software solution designed to handle these operations without failure. Information from many sources shows that these systems help to improve productivity, reduce errors, improve customer satisfaction, and help in decision making. This chapter shows different studies related to how to implement HMS, identifying gaps, and providing information that justify the development of a better and efficient hotel management system.

2.2 Conceptual Review

The conceptual review of a Hotel Management System is shown by this components, including reservation management, room allocation, billing, customer data storage, and making reports. According to Laudon and Laudon (2020), a management information system such as HMS is designed to provide information to support decisions in most functional areas. In the case of hotels, this involves providing real-time room availability, managing guest information, and automating the billing process.

The system's functionality is mostly supported by databases, user interfaces, and network connections that ensure good information sharing. Things such as database information, real-time processing, and user security by authentication are important for ensuring data integrity and security. Other sources such as O'Brien and Marakas (2019) emphasize the use of user-friendly interfaces and the ability of the system to expand to adapt if the demand is needed. Furthermore, modern HMSs interact with

platforms such as online booking systems and payment systems that improve how they are used and make them easy to access.

A Software Development Life Cycle (SDLC) provides a guide for creating a better HMS, from requirement analysis through design, implementation, testing, deployment, and maintenance. A conceptual understanding of this cycle makes sure that the system works with both user needs and organization needs.

2.3 Empirical Review

Many studies support the use of Hotel Management Systems in real-world applications. For example, a study conducted by Mohammed and Ismail (2018) on hotel automation in Nigerian urban areas showed that the introduction of a computerized system reduced customer check-in time and made booking more easy. Similar, research by Wang et al. (2017) in China showed that most small and medium-sized hotels that adopted HMS recorded a 25% increase in customer intake due to improved service delivery.

In Kenya, a case done by Ndungu and Otieno (2021) focused on budget hotels in Nairobi showed that most hotels had digital front-desk solutions, but only a few had automated billing and reporting systems. The lack of a fully digitized system led to delays and inaccurate financial problem, especially the seasons where the customers are many. This shows there is need to develop a system that can accommodate all the needed features.

Another study by Singh (2016) in India talked about the role of real-time analytics in improving customer satisfaction. The study showed that hotels using systems that could track customer preferences and behaviors were helping in providing more better services that is fit for most customers. These findings show that using features like customer history tracking,

feedback recording, and customized service notifications can improve customer satisfaction hence better results.

From this studies, it shows that while existing systems provide support to hotel operations, most have issues of poor integration, less accessibility, or less-user-friendly interfaces. This shows the need for an HMS that is not only good in operations but also good to use remotely and in real-time, especially in countries which are growing digitally like Kenya.

2.4 Conclusion of Literature

In conclusion, the study provides theory and practical study that helps understand the importance and structure of Hotel Management Systems. An HMS must support all important hotel functions while being easy to expand, more secure, and user-friendly. More studies show that the benefit of the system brings hotel operations globally and even country wide, while also showing weaknesses in existing systems. These study shows the need of developing , accessible, and efficient hotel management system that solves the challenges faced by many local hotels in Kenya.

CHAPTER THREE: WORK PLAN

3.1 Introduction

This chapter shows the preparation, planning and development of the proposed Hotel Management System. It includes feasibility analysis, requirement gathering, system design, and a schedule to guide the project. The aim is to ensure that the system is practical, functional, and easy to implement within the specified time and environment.

3.2 Feasibility Study

The feasibility study helps to determine whether the proposed system can be developed and used effectively. They are as follow, technical, economic, operational, and legal feasibility evaluations.

3.2.1 Technical Feasibility

The proposed system can be done in technical feasibility. Most hotels have access to basic IT infrastructure such as computers, and internet that is needed to run a web system or desktop system for Hotel Management. The project will be developed using technologies such as PHP, MySQL, and HTML/CSS for front-end, which are mostly accessible, cheaper, and supported by most devices.

3.2.2 Economic Feasibility

The system can be done in economic feasibility, as it reduces costs which are long term such as labor, paper usage, and human errors. The developed system will mainly be free and easy to use software, reducing licensing costs. Implementation will lead to increased in revenue and work will be done more efficiently through improved service delivery and financial accuracy.

3.2.3 Operational Feasibility

The system will make it easy to do booking, billing, kitchen orders, and guest data management, making it highly needed for hotel operations. Staff can be easily trained to use the system since it is user friendly, hence showing operational feasibility can be done.

3.2.4 Legal Feasibility

The system will comply with Kenya's data protection laws and ICT regulations by also using features that secure customer information, especially during money transactions. All data processes will follow the standard cybersecurity protocols.

3.3 Functional Requirements

The functional requirements shows important tasks the system must be able to perform:

- Allow customers to book rooms online or at the reception desk.
- Enable the reception to manage check-in and check-out digitally.
- Track food and drinks orders in real time.
- Generate accurate and secure invoices and receipts.
- Provide a secure database for customer and transaction records.
- Allow hotel managers to view reports on bookings, income, and service usage on real-time basis.

3.4 Non-Functional Requirements

These define how the system will perform its functions such as; Security: the system must secure all customer data and financial records. Usability: The interface must be simple and user friendly. Reliability: The system should function well without crashing. Scalability: The system should allow adding of more features and any system improvement needed. Performance: Response times should be fast, even with multiple concurrent users. Tools and Technologies to be Used Frontend are as follows; HTML, CSS, JavaScript Backend: PHP Database: MySQL Development Environment: XAMPP or WAMP server. Other Tools: Visual Studio Code, Git for version control, and Postman for API testing

3.6 Project Schedule

Table 1

Activity	Duration	Timeline
Requirements gathering	1 week	Week 1
System design (UML, wireframes)	1 week	Week 2
Backend and frontend development	3 weeks	Weeks 3 – 5
Integration and testing	2 weeks	Weeks 6 – 7
User training and documentation	1 week	Week 8
Final testing and deployment	1 week	Week 9

3.7 Expected Challenges

- Possible delays in requirements mostly due to stakeholders not being available.
- Hotel staff resisting change since they were used to the old system.
- Ensuring system security and compliance with data privacy laws

- It will be time consuming to make the system.

3.8 Mitigation Strategies

- Schedule regular meetings with stakeholders for clarity and feedback
- Do early training sessions with the staff so that they can familiarize with the system.
- Apply encryption and authentication to data for security
- Follow the project schedule strictly so as to use less time.